A Data-Centric Pipeline for Microscale Insect Detection in Industrial Sticky Traps

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Abstract. Industrial pest control is critical to food safety yet remains largely manual, with inspectors visually counting insects in sticky traps. This process is slow, expensive, non-scalable, and prone to high human error. Automating this task via Computer Vision faces severe small object detection challenges, such as the obliteration of morphological details due to feature vanishing, severe background clutter from industrial debris mimicking biological targets, and specimen occlusion caused by natural crowding on the adhesive surface. The literature presents two competing approaches: model-centric methods enhancing architecture and volume-centric methods expanding data through synthetic augmentation. However, neither systematically evaluates the impact of capture environment standardization on detection performance. This work proposes a data-centric pipeline prioritizing physical capture control, dataset curation, and ontological refinement. Using an identical YOLO architecture across all experiments, we evaluate the evolution of the Sticky Trap Insect Dataset family across its distinct curation stages. Results show mAP@0.5 improvements from 0.476 to 0.608 on the smallest model variant (YOLO26-S) achieved exclusively through data engineering, providing robust empirical evidence that capture standardization and dataset quality are the primary performance bottlenecks in microscale object detection, fundamentally superseding architectural innovation.

Keywords: Pest Control · Computer Vision · Small Object Detection

1 Introduction

Pest control in food-processing industries is strictly regulated by frameworks like ISO 22000:2018 [1] and Hazard Analysis and Critical Control Points (HACCP) [2].

Compliance currently relies on the manual inspection of sticky traps [3], a slow, non-scalable process with error rates up to 30% due to cognitive fatigue [4]. While Computer Vision offers an automated alternative, it faces severe Small Object Detection (SOD) challenges because the targets are ultra-small. Primary obstacles include "feature vanishing" during network downsampling [5], background clutter from adhesive residues that visually mimic insects, and specimen occlusion caused by natural crowding on the traps.

The literature addresses SOD through three competing approaches. Model-centric methods propose increasingly sophisticated architectures: lightweight YOLO variants optimized for embedded inference [6], NMS-free training combined with SPD-Conv modules to recover spatial information lost in strided convolutions [7], and specialized CNNs capable of simultaneously classifying species, sex, and physiological status [8]. Volume-centric methods, in contrast, attempt to compensate for data scarcity through synthetic image generation—typically via 3D rendering, or through aggressive augmentation pipelines. A third, inference-centric line of work, exemplified by SAHI [9], intervenes neither on the training data nor on the architecture, but on the inference pipeline itself, slicing high-resolution images into overlapping patches to preserve small-object features that would otherwise be lost during standard convolution. None of these three approaches, however, addresses the root cause of the problem: the procedure in which training images are acquired. This gap motivates the central questions driving our work:

RQ1: To what extent does dataset quality and physical capture standardization outperform increased model complexity in microscale insect detection?

RQ2: Which component of a data-centric pipeline (ontological refinement, physical capture standardization, or iterative curation) yields the greatest individual performance gain?

RQ3: Can a low-complexity model, trained on a rigorously curated dataset, achieve industrially acceptable detection performance without architectural modification?

Our results establish the following contributions:

- A reproducible, three-component data-centric pipeline for microscale object detection in industrial environments.
- The Sticky Trap Insect Dataset (STID) dataset family, four progressively curated versions providing controlled ablation of each pipeline stage.
- Empirical evidence that data engineering alone improves mAP@0.5 from 0.476 to 0.608 on a fixed YOLO baseline (YOLO26-S) [10], demonstrating that capture standardization supersedes architectural complexity in this domain.

2 Related Work

The dominant line of research in pest detection attributes performance limitations to architectural deficiencies and responds by proposing increasingly sophisticated neural networks. The YOLO family remains the most widely adopted backbone for real-time detection tasks, and several domain-specific variants have been developed. YOLO-DCPG [6] introduces a lightweight modification of YOLOv8n aimed at embedded deployment, reducing parameter count by over 50% while improving precision on pest imagery. YOLO-YSTs [7] combines the NMS-free training strategy of YOLOv10 with SPD-Conv modules to mitigate information loss in strided convolutions, a critical issue for millimeter-level targets. VectorBrain [8] extends the task further, concurrently predicting species identity, sex, and abdomen status from a single forward pass.

While these contributions are technically significant, they share a common assumption: that the training data is of sufficient quality and consistency to allow the model to learn. A second body of work attributes performance limitations to data scarcity and attempts to address it through volume expansion. Synthetic image generation—typically via 3D rendering engines [11], allows the creation of large annotated datasets without manual collection. Augmentation pipelines further multiply existing samples through geometric and photometric transformations [12].

A growing body of work challenges the model-centric consensus by arguing that, for many tasks, dataset quality is the primary determinant of performance. Ng [11] formalizes this position under the **Data-Centric AI** paradigm, demonstrating that systematic dataset improvement yields larger performance gains than architectural iteration. In the SOD literature, SAHI [9] proposes a data-pipeline intervention at inference time, slicing images into overlapping patches to preserve spatial features typically lost in standard convolutions. Research on edge computing for vision-based insect traps [13] further indicates that unstandardized datasets are a primary cause of poor field performance, and that physical controls on image acquisition can significantly improve model reliability.

As Table 1 demonstrates, no existing work applies a complete data-centric pipeline to microscale industrial pest detection. While current literature predominantly relies on a model-centric paradigm, our work uniquely integrates a physical Standard Operating Procedure (SOP) to prevent specimen occlusion and targeted Noise Mitigation to resolve background clutter. Most importantly, we validate this pipeline through rigorous data ablation, isolating the impact of data maturation on a fixed architecture to prove that dataset structural integrity—not algorithmic complexity, is the true bottleneck in this domain.

3 Methodology

3.1 Experimental Setup

All experiments were conducted using the YOLO26 detection framework [10] in its S, M, L, and X size variants. Relative to prior Ultralytics releases, YOLO26

Table 1. Comparison of related work by methodological focus.

Work	Core Paradigm	Physical SOP	Noise Mitigation	Data Ablation
YOLO-DCPG [6]	Model-centric	×	×	×
YOLO-YSTs [7]	Model-centric	×	×	×
VectorBrain [8]	Model-centric	×	×	×
SAHI [9]	Inference	×	×	×
Silva et al. [13]	Data-centric	Partial	×	×
Ours (STID)	**Data-centric**	✓	✓	✓

adopts a natively end-to-end, NMS-free detection head, removes Distribution Focal Loss in favor of direct coordinate regression, and is trained with the MuSGD optimizer alongside a Small-Target-Aware Label assignment strategy aimed at improving small-object recall, making it a relevant and representative architecture for this study. The complete set of hyperparameters and hardware specifications is detailed in Table 2. The input resolution was fixed at native Full HD (1920×1080 pixels) to preserve the millimetric morphological details of the insect targets, which required a batch size of 1 to avoid out-of-memory errors on the larger variants. The dataset was partitioned into training, validation, and test sets following an 80%/10%/10% split, applied consistently across all STID versions to prevent data leakage.

Table 2. Training hyperparameters and experimental environment.

Parameter / Component	Value / Specification
Epochs	200
Batch Size	1
Input Resolution	1920×1080 pixels
Optimizer	SGD
Initial Learning Rate (lr_0)	0.01
Data Split (Train/Val/Test)	80% / 10% / 10%
Compute Node	NVIDIA DGX V100 System
CPU	Intel Xeon
Framework	PyTorch

3.2 Dataset Curation and Baseline (STID-Clean)

The initial phase of the pipeline utilized STID-Clean, consisting of 500 original captures of adhesive traps collected under uncontrolled field conditions. This baseline allowed for the systematic identification of real-world challenges inherent to microscale object detection, including *feature vanishing* during network downsampling, severe background clutter caused by industrial residues, and abrupt scale variations arising from inconsistent camera distances. STID-Clean was annotated with five biological classes, namely Fruit Fly (FF), House Fly (HF), Blow Fly (BF), Flesh Fly (MF), and a generic Other Insects (OI) class covering specimens that did not fall into the four named taxa, and served as the reference point against which all subsequent curation stages were evaluated. All images were captured at a fixed resolution of $1{,}920 \times 1{,}080$ pixels and annotated using Roboflow, with tight axis-aligned bounding boxes drawn to the morphological

limits of each specimen. From STID-Onto onward, a sixth, non-biological class, *Noise*, was added to explicitly annotate industrial debris (adhesive dirt, fibres, and particulate matter) that visually resembles the biological targets, as detailed in Section 3.4.

3.3 Volume Expansion via Synthetic Data (STID-Raw)

To address data scarcity, a second dataset version, STID-Raw, was produced by augmenting the original captures with synthetic images generated in Blender. The 3D modeling pipeline randomized the spatial distribution and angular rotation of virtual insect models, positioning each specimen at the exact focal depth of the adhesive layer to maximise photorealism. This process yielded over 2,700 training samples. Despite the increased volume, the approach was ultimately discarded: artificial lighting conditions and the absence of organic adhesive textures prevented the model from generalizing to real trap images, confirming that volume alone is insufficient when domain fidelity is low.

3.4 Ontological Refinement (STID-Onto)

Retracting from the synthetic approach, the third iteration refocused on annotation quality. STID-Onto introduced a formal ontological refinement by expanding the label set from five to six classes through the addition of an auxiliary *Noise* class. This class was specifically designed to encode the visual appearance of industrial debris, adhesive dirt, fibres, and particulate matter, that consistently triggered false positive detections when morphologically similar to the Fruit Fly. By explicitly forcing the network to learn what *not* to detect, this intervention significantly reduced the false positive rate in real operating conditions. Data augmentation was applied exclusively to the organic captures at this stage to further improve sample diversity without reintroducing the domain gap observed in STID-Raw.

3.5 Physical Capture Standardisation (STID-SOP)

The final and most impactful stage of the pipeline addressed the physical bottlenecks identified in STID-Clean: inconsistent focal distance, variable lighting, and specimen occlusion due to natural crowding on the adhesive surface. A rigorous SOP was established, fixing the camera focal length and enforcing uniform illumination across all acquisitions, thereby guaranteeing a constant spatial resolution at the millimetre scale. In parallel, structured physical templates, printed grids placed directly on the adhesive plates—imposed a homogeneous spatial distribution of specimens, effectively eliminating occlusion and enabling bounding-box annotations to be adjusted strictly to the morphological limits of each individual.

We refer to the cumulative set of real, non-synthetic captures accumulated across the STID-Clean and STID-Onto stages as the *organic dataset*, as opposed to the Blender-generated synthetic images introduced in STID-Raw (Section 3.3). STID-SOP is composed exclusively of images captured under these

standardized 2026-protocol plates. Subsequent internal versions within the 5th iteration series combine these standardized captures with selected legacy images from the organic dataset, applying class balancing to mitigate the natural imbalance between abundant and rare species.

3.6 Dataset Summary

Table 3 summarizes the four STID versions, their core engineering strategy, and approximate sample counts. Together, they form a controlled ablation of the proposed pipeline.

Table 3. Evolution of dataset versions and engineering strategies.

Version	Approx. Samples	Key Engineering Strategy
STID-Clean	∼500	Organic baseline; 5 biological classes; uncontrolled capture conditions.
STID-Raw	>2,700	Volume expansion via Blender 3D synthetic images; randomised specimen placement and rotation.
STID-Onto	∼500	Ontological refinement; 6 classes; *Noise* class introduced to suppress false positives from industrial debris.
STID-SOP	∼900	Physical SOP: fixed focal length, uniform lighting, printed grid templates to eliminate specimen occlusion.

To validate the three research questions, we trained YOLO26 in four size variants (S, M, L, and X) on each of the four STID dataset versions, yielding a factorial grid of $4 \times 4 = 16$ experimental conditions. Performance was measured by Mean Average Precision at Intersection over Union (IoU) 0.5 (mAP@0.5), Precision, and Recall. Holding the architecture and training configuration constant across all runs ensures that any observed variation is attributable exclusively to changes in the underlying data, providing the controlled ablation necessary to answer each research question. The STID dataset was collected under a commercial partnership and contains proprietary industrial data; consequently, it is not publicly released, though anonymized samples may be made available upon reasonable request for academic verification purposes.

4 Experiments

Due to computational constraints imposed by the fixed Full HD resolution and unitary batch size, each configuration was trained with a single run rather than averaged across multiple random seeds; we report point estimates and note this as a limitation to be addressed in future replications of this study.

4.1 RQ1 — Data Quality versus Model Complexity

Reading Table 4 *horizontally* (across model scales within a fixed dataset) reveals that scaling from YOLO26-S to YOLO26-X yields negligible and inconsistent im-

Table 4. Detection performance (mAP@0.5 / Precision / Recall) across dataset versions and model scales.

Dataset	YOLO26-S			YOLO26-M			YOLO26-L			YOLO26-X		
	mAP	P	R	mAP	P	R	mAP	P	R	mAP	P	R
STID-Clean	0.476	0.423	0.625	0.452	0.368	0.545	0.482	0.430	0.592	0.481	0.449	0.589
STID-Raw	0.679	0.676	0.652	0.675	0.665	0.632	0.677	0.643	0.644	0.645	0.628	0.641
STID-Onto	0.381	0.574	0.504	0.415	0.404	0.554	0.402	0.375	0.595	0.409	0.415	0.560
STID-SOP	0.608	0.527	0.652	0.492	0.592	0.522	0.547	0.461	0.583	0.534	0.455	0.549

provements. On STID-Clean, mAP@0.5 ranges from 0.476 to 0.481, an absolute spread of only 0.005 points despite a substantial increase in parameter count. The same plateau appears on STID-Onto (0.381–0.409) and STID-SOP (0.608–0.534). Across every dataset version, the gain from multiplying model capacity is marginal and does not follow a consistent upward trend.

Reading the table *vertically* (across dataset versions within a fixed model) reveals the opposite dynamic. Even the smallest variant, YOLO26-S, improved its mAP@0.5 from 0.476 on STID-Clean to 0.608 on STID-SOP, an absolute gain of +0.132. This pattern is consistent across all four model sizes, suggesting that the performance bottleneck resides in the data rather than in the network architecture.

The explanation lies in the physical nature of the task. When training images contain occluded or scale-inconsistent specimens, the morphological features required for detection are absent or corrupted at the pixel level. No increase in model capacity can reconstruct information that was never captured at image acquisition. Once physical controls were applied in STID-SOP, the smallest YOLO26-S architecture learned to detect the same targets that larger models consistently failed on, confirming that architectural complexity cannot compensate for spatially deficient input.

4.2 RQ2 — Contribution of Each Pipeline Component

The vertical trajectory of Table 4 reveals three qualitatively distinct phases of dataset evolution, each corresponding to a specific pipeline intervention.

Phase 1: Volume expansion (STID-Raw). The STID-Raw dataset was constructed by merging the original 500 real captures with more than 2 ,700 synthetic images generated by Blender 3D modeling, following the volume-centric hypothesis that more training data would benefit the model. The validation metrics for STID-Raw, to be fully reported upon completion of current training runs, are expected to confirm an apparent improvement over STID-Clean on the held-out validation split.

However, this apparent gain is accompanied by a qualitative failure mode observed during visual inspection of predictions on genuine adhesive plates outside the training distribution: models trained on STID-Raw exhibited a higher rate of false negatives on real specimens and false positives on adhesive residue, consistent with a domain gap between the synthetic and organic image distributions. Synthetic images, despite correct geometric placement, lack the organic surface

textures of real insect bodies and the characteristic appearance of adhesive substrates under authentic illumination conditions. A quantitative evaluation on held-out real-world deployment data is left for future work.

This phenomenon illustrates a fundamental limitation of volume-centric approaches: validation-set performance computed on in-distribution data (where synthetic and real are mixed) can be a misleading proxy for real-world reliability when the training distribution diverges from the deployment distribution. Notably, STID-SOP relies exclusively on real captures under standardized conditions, which by construction avoids this synthetic-to-real domain gap.

This finding demonstrates that, for microscale biological targets with distinctive organic textures, synthetic data augmentation not only fails to generalize but actively misleads evaluation, creating a false impression of progress that disappears under real operational conditions.

Phase 2: Ontological refinement (STID-Onto). Recognizing that texture fidelity alone was insufficient, STID-Onto introduced an ontological intervention: the addition of an auxiliary *Noise* class comprising annotated industrial debris and adhesive residue that visually mimics millimeter-sized insects. This forced the network to explicitly learn what *not* to detect, providing direct supervision over the primary source of false positives. The resulting mAP@0.5 for YOLO26-S reached 0.381, lower in absolute terms than STID-Clean, but accompanied by substantially higher Precision (0.574 versus 0.423), indicating that the model began making fewer false biological detections.

Phase 3: Physical capture standardization (STID-SOP). The decisive intervention was the implementation of a Standard Operating Procedure (SOP) that fixed focal length and standardized illumination, combined with printed grid templates on the adhesive plates to enforce specimen separation. This addressed the two physical bottlenecks that ontological refinement alone could not solve: inter-image scale variance and bounding-box overlap from specimen crowding. The transition from STID-Onto to STID-SOP produced the largest single gain across all model scales, with YOLO26-S reaching a peak mAP@0.5 of 0.608, an absolute improvement of +0.227 over STID-Onto. Crucially, unlike STID-Raw, these metrics reflect real, standardized captures rather than a mix including synthetic imagery, confirming that physical standardization produces a training distribution that genuinely matches the deployment environment.

4.3 RQ3 — Industrial Viability of a Standard Model

The third research question asks whether an architecturally unmodified model, trained on a rigorously curated dataset, can achieve detection performance comparable to that reported by specialized architectures in the literature.

The results confirm this hypothesis. YOLO26-S trained on STID-SOP achieved a global mAP@0.5 of 0.608, with Precision of 0.527 and Recall of 0.652. These figures are directly comparable to those reported by architecturally customized models such as YOLO-DCPG [6] and YOLO-YSTs [7], which operate on less challenging, better-controlled datasets and rely on structural modifications to achieve their gains. Our baseline model, without any such modifications, reaches

equivalent performance through data quality alone. Notably, on our own dataset this relationship inverts once the data-centric pipeline is applied: on STID-Clean, YOLO26-L holds a marginal edge over YOLO26-S (0.482 versus 0.476), whereas on STID-SOP YOLO26-S (0.608) clearly outperforms YOLO26-L (0.547). This reversal is direct evidence that dataset quality, not model capacity, is the dominant factor governing detection performance in this task.

Furthermore, considering the benchmark established by manual inspection (the Technique for Human Error Rate Prediction, THERP, estimates human error rates of up to 30% in repetitive visual tasks [4]), a Recall of 0.652 on an unconstrained microscale dataset represents a performance level that is industrially competitive. As physical standardisation is progressively extended to the full operational environment, these metrics are expected to improve further.

The residual classification errors provide additional evidence that the remaining limitations are physical rather than algorithmic. The confusion between Flesh Fly (MF) and Blow Fly (BF) reflects genuine morphological similarity that challenges even trained human taxonomists. The misclassification between Fruit Fly (FF) and the *Noise* class occurs because sub-millimeter debris is physically indistinguishable from the insect's body under standard imaging conditions. These are not model failures; they are the irreducible boundaries of the detection task itself.

5 Conclusion

This work demonstrates that, for the task of microscale object detection in industrial pest control, the structural integrity of the training dataset is the primary determinant of model performance, fundamentally superseding architectural complexity. Through a controlled ablation across four progressively curated versions of the STID dataset and four YOLO26 model scales, we showed that scaling model capacity within a fixed dataset yields negligible gains, whereas systematic data engineering, comprising ontological refinement through the introduction of a *Noise* class and physical capture standardisation via a printed-grid SOP and fixed focal distance, produced the decisive performance gains observed. The residual classification errors in the final model, such as the confusion between morphologically similar species, reflect the taxonomic boundaries of the target species rather than algorithmic deficiencies, suggesting that scaling the pipeline to a larger number of visually similar species will likely require finer-grained morphological features rather than dataset volume alone. Dense crowding scenarios beyond what the printed-grid SOP resolves may also benefit from instance segmentation, though this was not explored here given its added annotation cost. Future work will focus on deploying the model to edge devices for real-time field monitoring, validating the pipeline across multiple random seeds, and investigating self-supervised learning techniques to reduce the manual annotation effort required for emerging pest species.

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